

Week 18. Industry Synthesis

Documentation & Final Presentations

- Bringing it all together; From technical delivery to professional handover
- Demonstrating end-to-end ML, scripting, and cloud workflows
- Preparing for real-world integration; Ready for workplace expectations

Learning Objectives & Session Agenda

- Summarise skills in ML, scripting, and cloud workflows in a production pipeline
- Prepare professional handover documentation for your project
- Present distributed training results to a technical audience
- Reflect on responsible AI practices in workplace scenarios
- Agenda; Documentation standards, Presentations, Responsible AI, Assessment wrap-up

The Role of Documentation & Handover in Industry

- Essential for continuity and project success in ML and AI teams
- Enables maintainability and knowledge transfer across teams
- Helps organisations reduce risk and support production deployments

Documentation Standards in ML Projects

- Use clear, concise language explaining code, pipelines, and deployment processes
- Include system diagrams, UML, flowcharts, and data schemas where needed
- Integrate experiment tracking, reproducibility steps, and environment details

Workplace Handover; Example Structure

- Executive summary and solution overview
- Code repository links, environment config, and requirements as lists
- How to run, train, and test models step by step
- Troubleshooting, tips, and points of contact for further support

Reporting & Presenting to Stakeholders

- Translate technical details into value for business and technical audiences
- Utilise graphs, tables, and simple performance metrics for clarity
- Prepare a professional portfolio including project docs, visuals, and reflections

Responsible AI; Team and Stakeholder Reflections

- Document decision areas relating to fairness, reliability, and transparency
- Include risk assessments, bias identification, and mitigation strategies
- Provide recommendations for ethical operations and future monitoring

Final Presentations; Requirements and Expectations

- Present distributed training project; Focus on workflow, results, and learnings
- Address performance against industry benchmarks and MLOps practices
- Demonstrate understanding of workplace readiness through clear communication

Assessment Checklist & Submission

- All components must be submitted; code, documentation, and presentation
- Follow minimum requirements; clear, complete, and reproducible assets
- Use provided templates and checklists to ensure workplace standard submission

Activity; Team Review & Presentation

- In groups, review final project docs against checklist for clarity and completeness
- Each student presents project highlights; Explain technical approach to layperson and technical peer
- Peer feedback session; Rate clarity, professionalism, and responsible AI considerations

Workforce Reflection; Preparing for Next Steps

- Reflect on how ML, cloud, and scripting integration can support industry demands
- Identify your strengths and areas for continued learning; Set goals for the next 6 months
- Consider ways to apply responsible AI and documentation best practices in your first job role

Summary & Next Steps

- Recap skills; PyTorch, scripting, cloud integration, ML engineering
- Emphasise importance of clear docs, responsible AI, and professional delivery
- Reassessment window follows; Final queries and support available in weeks 19-20

